

MA540 Project 3

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March 2026

Problem 1

Statement

We analyze the steady state heat model for a copper rod, which follows the boundary value problem

$$\frac{d^2 T_s}{dx^2} = \frac{2(a+b)}{ab} \frac{h}{k} [T_s(x) - T_{amb}]$$
$$\frac{dT_s}{dx}(0) = \frac{\Phi}{k} \quad , \quad \frac{dT_s}{dx}(L) = \frac{h}{k} [T_{amb} - T_s(L)]$$

We can use the known thermal conductivity $k = 4.01 \frac{W}{cm \cdot ^\circ C}$, and we take $u_{amb} = 22.28$. The cross-sectional dimensions of the rod are $a = b = 0.95$ cm with length $L = 70$ cm. We will use the data in Table 3.3:

x (cm)	10	14	18	22	26	30	34	38
Temp ($^\circ C$)	66.04	60.04	54.81	50.42	46.74	43.66	40.76	38.49
x (cm)	42	46	50	54	58	62	66	
Temp ($^\circ C$)	36.42	34.77	33.18	32.36	31.56	30.91	30.56	

Here we will construct densities for Φ and h using both the random walk Metropolis Algorithm (MCMC), with $M = 10^4$ iterations, and the delayed rejection adaptive Metropolis Algorithm (DRAM). We will employ the covariance estimate found from OLS.

$$V = \begin{bmatrix} 8.964 * 10^{-3} & -1.212 * 10^{-6} \\ -1.212 * 10^{-6} & 1.7726 * 10^{-10} \end{bmatrix}$$

We will then compare our results with the frequentist estimates computed in Problem 1 of Project 2.

Solution

Figure 1 shows the marginal paths of each parameter under MCMC, and Figure 2 shows the marginal paths of each parameter under DRAM. We note that the mixing is similar for both, with minimal stagnation and high exploration. The chain covariance matrix, from DRAM, is

$$V = \begin{bmatrix} 9.8395 * 10^{-3} & -1.3457 * 10^{-6} \\ -1.3457 * 10^{-6} & 2.0038 * 10^{-10} \end{bmatrix}$$

which is close to the original OLS covariance matrix. The estimated standard deviations are $\sigma_\Phi = 9.9194 * 10^{-2}$ and $\sigma_h = 1.4156 * 10^{-5}$. The marginal density plots are observed in Figures 3 and 4. Note that approximately 95% of the density is within 2σ for both parameters.

Figure 1: Chains under MCMC

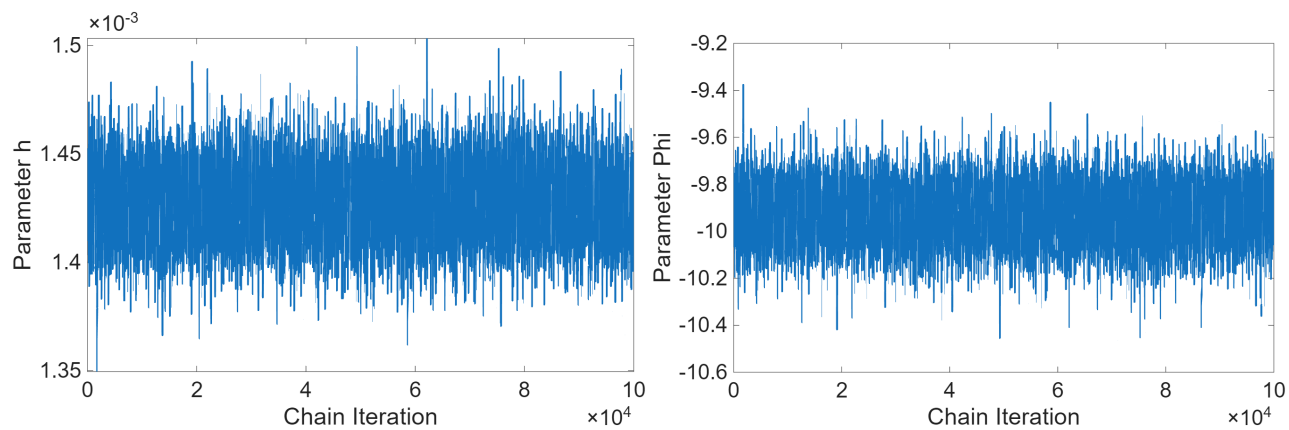


Figure 2: Chains under DRAM

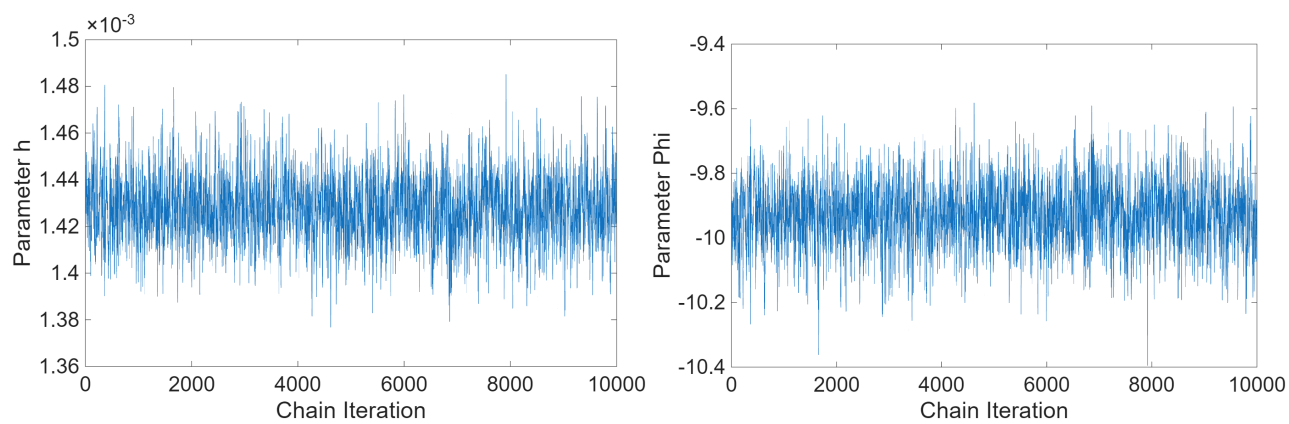


Figure 3: Marginal Densities under MCMC

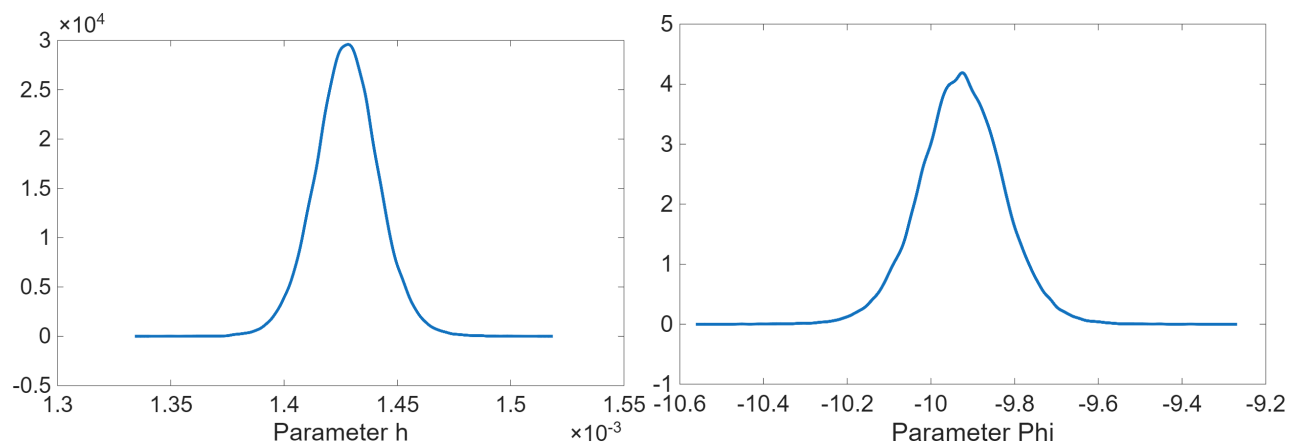
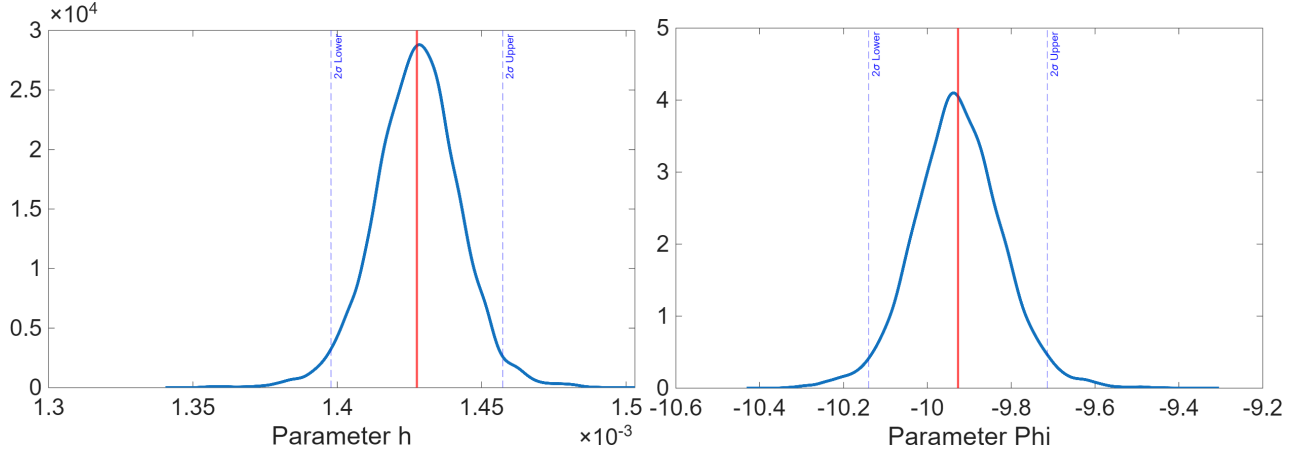


Figure 4: Marginal Densities under DRAM



In the below table, we compare the estimated standard deviations of error, Φ , and h by DRAM and the frequentist analysis in Problem 1 of Project 2:

σ	DRAM	Frequentist	% Difference
σ	$2.484 * 10^{-1}$	$2.301 * 10^{-1}$	0.76%
σ_{Φ}	$9.9194 * 10^{-2}$	$9.4678 * 10^{-2}$	4.65%
σ_h	$1.4156 * 10^{-5}$	$1.3314 * 10^{-5}$	6.189%

Problem 2a

Statement

We consider the SIR model

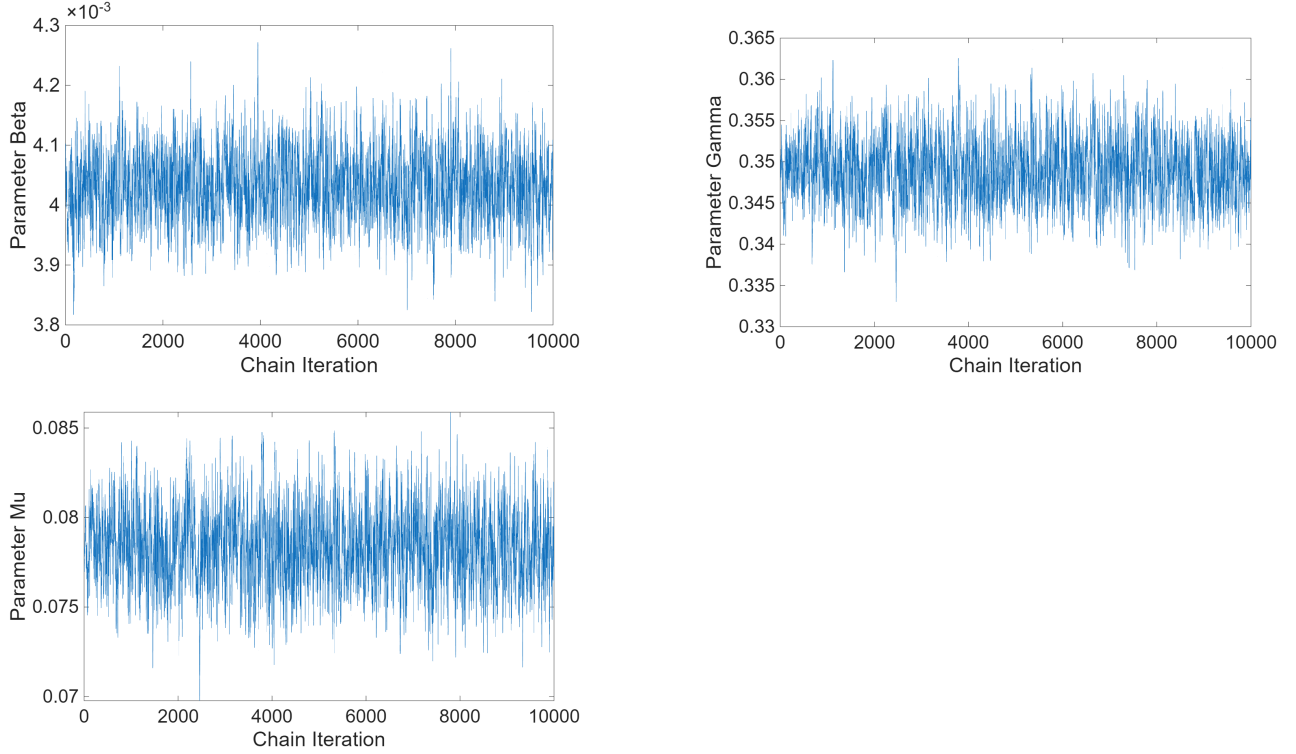
$$\begin{aligned}
 \frac{dS}{dt} &= \mu(N - S) - \beta IS \quad , \quad S(0) = 900, \\
 \frac{dI}{dt} &= \beta IS - (\gamma + \mu)I \quad , \quad I(0) = 100, \\
 \frac{dR}{dt} &= \gamma I - \mu R \quad , \quad R(0) = 0,
 \end{aligned}$$

where the parameters β, γ and μ are each in the interval $[0, 1]$. Here, we will use data from the file `SIR.integer.txt` to employ DRAM and compute chains, marginal densities, and pairwise plots for the parameters. We will also estimate the observations variance σ^2 . We will use the covariance matrix estimate from Project 2, Problem 3:

$$V = \begin{bmatrix} 4.96 * 10^{-10} & 7.05 * 10^{-10} & -7.14 * 10^{-10} \\ 7.05 * 10^{-10} & 2.21 * 10^{-6} & 9.27 * 10^{-7} \\ -7.14 * 10^{-10} & 9.27 * 10^{-7} & 6.85 * 10^{-7} \end{bmatrix}$$

We will then compare to OLS estimates.

Figure 5: 2a Chains under DRAM



Solution

Figure 5 shows the chains for each. We note that there is minimal stagnation and high exploration, so the chains converge. We estimate the observation variance $\sigma^2 = 4.6855 \times 10^2$. The OLS estimate of $\sigma^2 = 6.9092 \times 10^1$ differs greatly from this estimate, which indicates The mean parameter values found by DRAM are $\beta = 4.029 \times 10^{-3}$, $\gamma = 3.4891 \times 10^{-1}$, $\mu = 7.8397 \times 10^{-2}$, and the OLS estimates from `fminsearch.m` are $\beta = 4.0268 \times 10^{-3}$, $\gamma = 3.4875 \times 10^{-1}$, $\mu = 7.8226 \times 10^{-2}$, which are within 5%. The estimated variances found by DRAM are $\sigma_\beta = 5.7945 \times 10^{-5}$, $\sigma_\gamma = 3.9569 \times 10^{-3}$, $\sigma_\mu = 2.1763 \times 10^{-3}$, and the OLS estimates are $\sigma_\beta = 6.4358 \times 10^{-5}$, $\sigma_\gamma = 1.80 \times 10^{-3}$, $\sigma_\mu = 2.57 \times 10^{-2}$, which are outside the expected percent difference. We expect this is due to errors in Project 2 results. The final adapted covariance matrix is as follows:

$$V = \begin{bmatrix} 3.3576 \times 10^{-9} & 1.8531 \times 10^{-9} & -7.669 \times 10^{-9} \\ 1.8531 \times 10^{-9} & 1.5657 \times 10^{-5} & 6.6203 \times 10^{-6} \\ -7.669 \times 10^{-9} & 6.6203 \times 10^{-6} & 4.7364 \times 10^{-6} \end{bmatrix}$$

We note that this is similar to the initial estimate, but off by a power of 10.

Figure 6 shows the marginal densities for each parameter, graphed with the OLS distributions from Project 2. We note that the densities differ, but this is likely due to errors in the Project 2 results. The DRAM results are consistent, as for each parameter, 95% of the distribution is within 2σ intervals.

Figure 7 shows the pairwise plots for each parameter pair. We note that γ and μ appear to have some correlation, while the other pairs appear to be uncorrelated.

Figure 6: 2a Marginal Densities

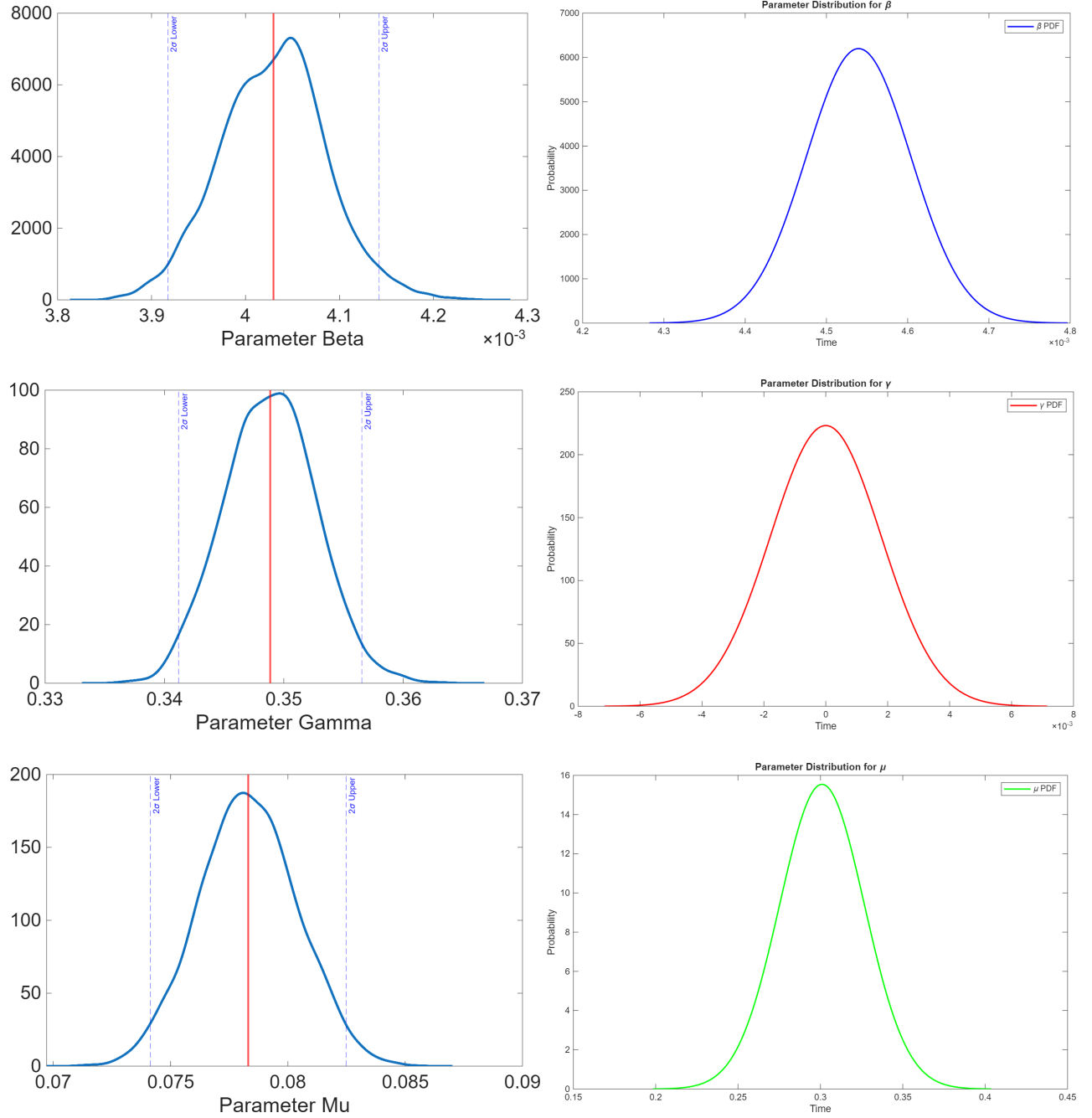
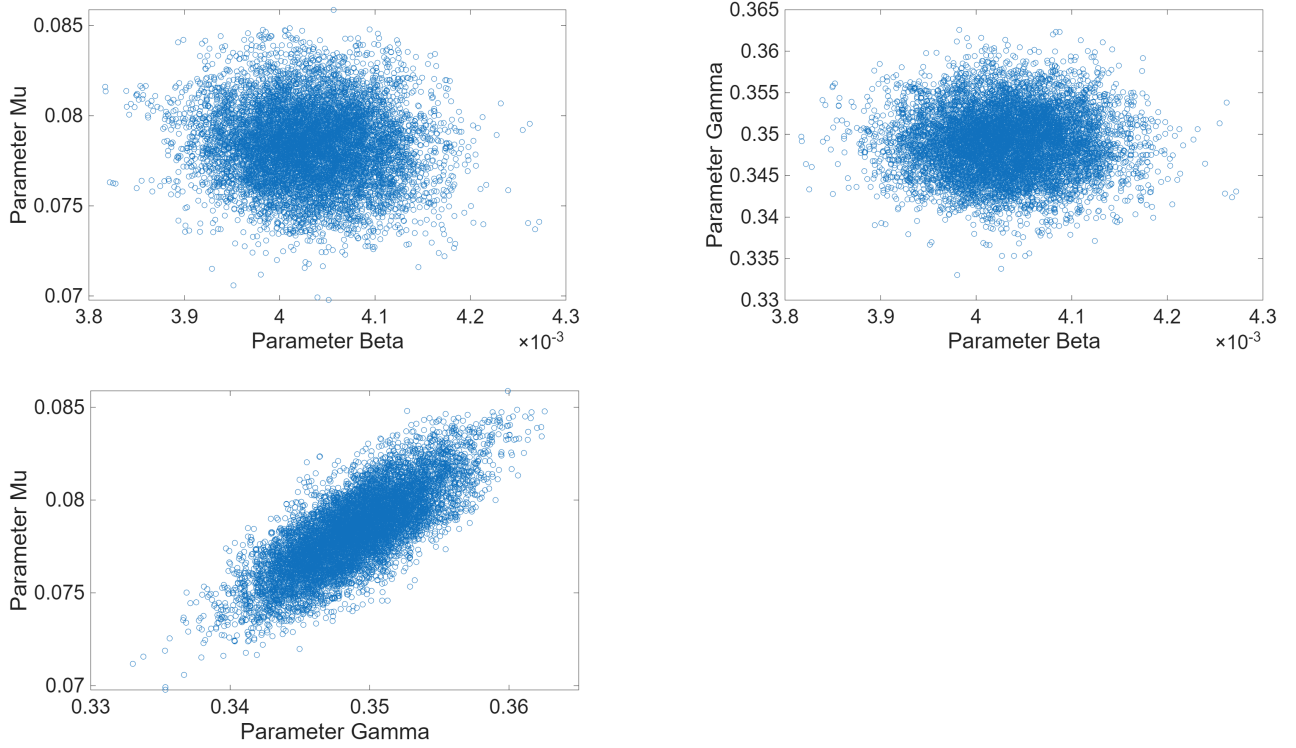


Figure 7: 2a Pairwise under DRAM



Problem 2b

Statement

We now consider

$$\begin{aligned} \frac{dS}{dt} &= \mu(N - S) - \eta k I S \quad , \quad S(0) = 900, \\ \frac{dI}{dt} &= \eta k I S - (\gamma + \mu) I \quad , \quad I(0) = 100, \\ \frac{dR}{dt} &= \gamma I - \mu R \quad , \quad R(0) = 0, \end{aligned}$$

with non-identifiable parameters $\theta = [\eta, \gamma, \mu, k]^T$. We will run DRAM and plot the pairwise distributions, and discuss the convergence of the chains.

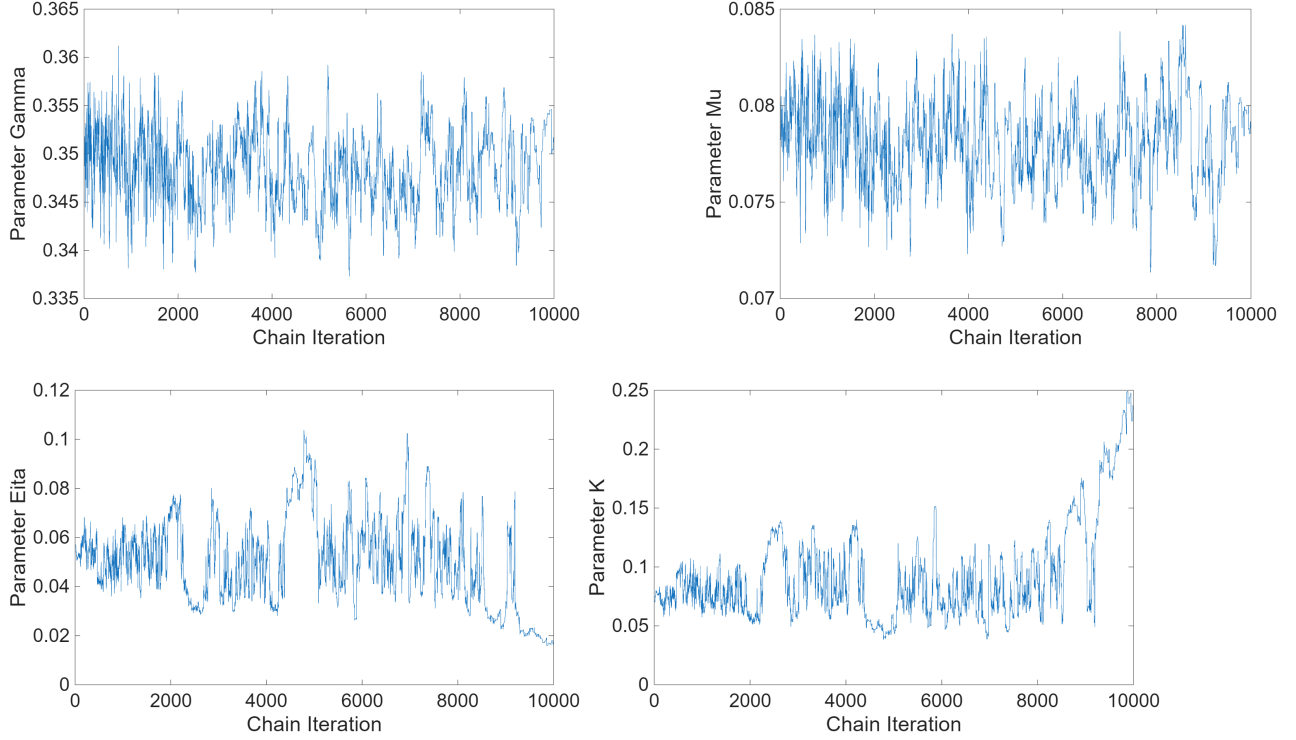
Solution

Figure 8 shows the chains of each parameter. Note that non-identifiable parameters η, k exhibit notable stagnation and diverge, while identifiable parameters γ, μ are shown to converge, with low stagnation and high exploration.

Figure 9 shows the distributions of each parameter. Note again the difference between the identifiable parameters, which follow normal distributions, and the non-identifiable parameters, which exhibit much more irregular behavior.

Figure 10 shows the pairwise plots of each parameter pair. Note the high correlation between η and k , which follows from their mutual non-identifiability. From these results, we see that Bayesian

Figure 8: 2b Chains under DRAM



analysis can help identify which parameters are non mutually identifiable from the divergence of their chains, as well as the irregular behavior of their estimated distribution.

Problem 2c

Statement

We consider again the model from Problem 2b, this time with the influenza data in Table 1: We

Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Confined to Bed	3	8	26	76	225	298	258	233	189	128	68	29	14	4

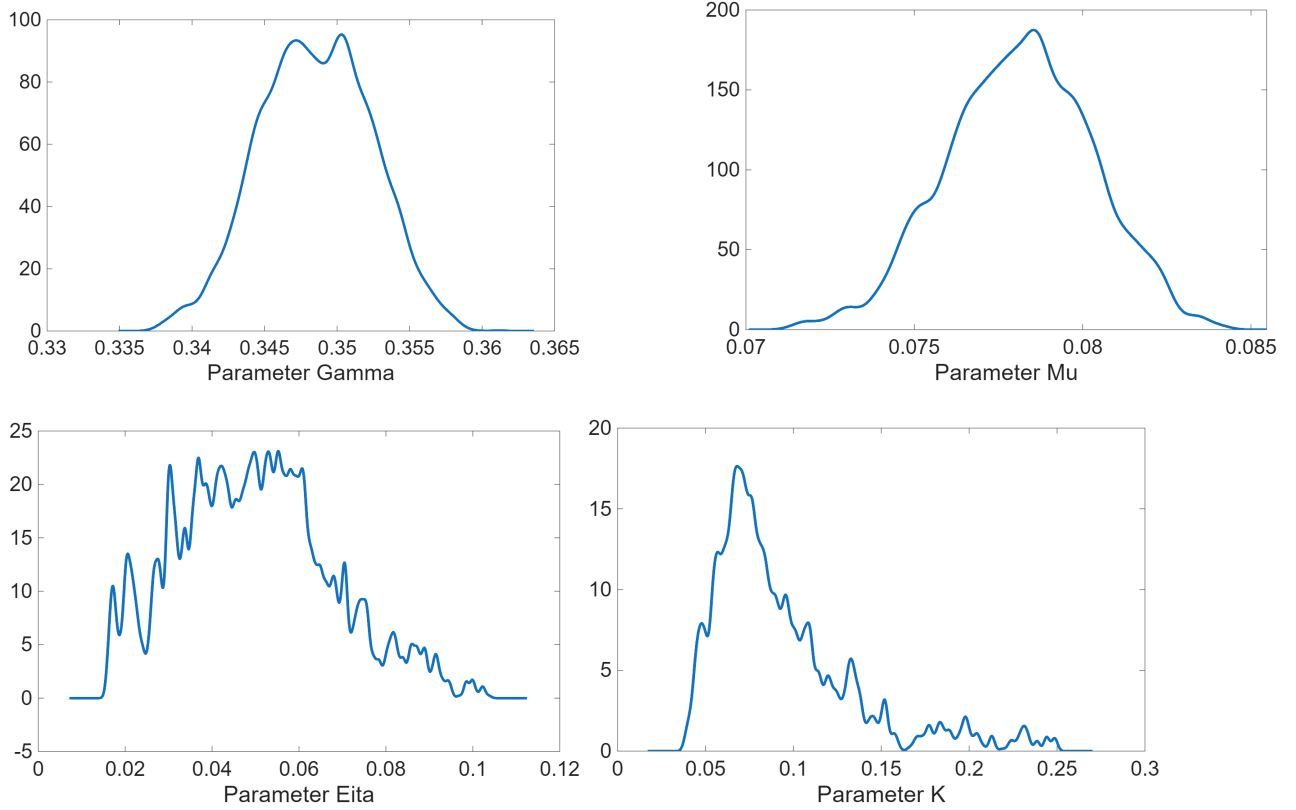
Table 1: Influenza data reported in [370] from the source [12].

specify $\mu = 0$ and take the initials $S(0) = 730$, $I(0) = 3$, and $R(0) = 0$, so $N = 733$. We will then use DRAM to construct distributions for β and γ , and compare the resulting mean values with the OLS estimates from Project 2.

Solution

Figure 11 shows the chains under DRAM, which converge with high exploration and low stagnation. Figure 12 shows the distributions for each parameter, which fit 95% within 2σ intervals. The mean parameter values are $\beta = 2.3078 \times 10^{-3}$, $\gamma = 4.3092 \times 10^{-1}$, which are within 5% of the OLS values of

Figure 9: 2b Densities under DRAM



$\beta = 2.3037 * 10^{-3}, \gamma = 4.2845 * 10^{-1}$. Figure 13 shows the pairwise plot, which indicates these two parameters are uncorrelated.

Problem 3

Statement

We will now analyze the Helmholtz energy

$$\psi(P, \theta) = \alpha_1 P^2 + \alpha_{11} P^4 + \alpha_{111} P^6,$$

using the data in the file `Helmholtz.txt` to employ both DRAM and MCMC, and compare the chains, marginal densities, and pairwise plots for the parameters in each. We will also report and compare the parameter estimates and observation variance, and discuss.

Solution

Figure 14 compares the chains by DRAM and MCMC for each parameter. We note high similarity and convergence for each chains. The parameter estimates by DRAM are $\alpha_1 = -374.41, \alpha_{11} = 712.52, \alpha_{111} = 211.64$, and by MCMC are $\alpha_1 = -373.53, \alpha_{11} = 707.14, \alpha_{111} = 218.20$. The observation variance by DRAM is $\sigma^2 = 12.910$, and by MCMC is $\sigma^2 = 12.737$.

Figure 10: 2b Pairwise Plots under DRAM

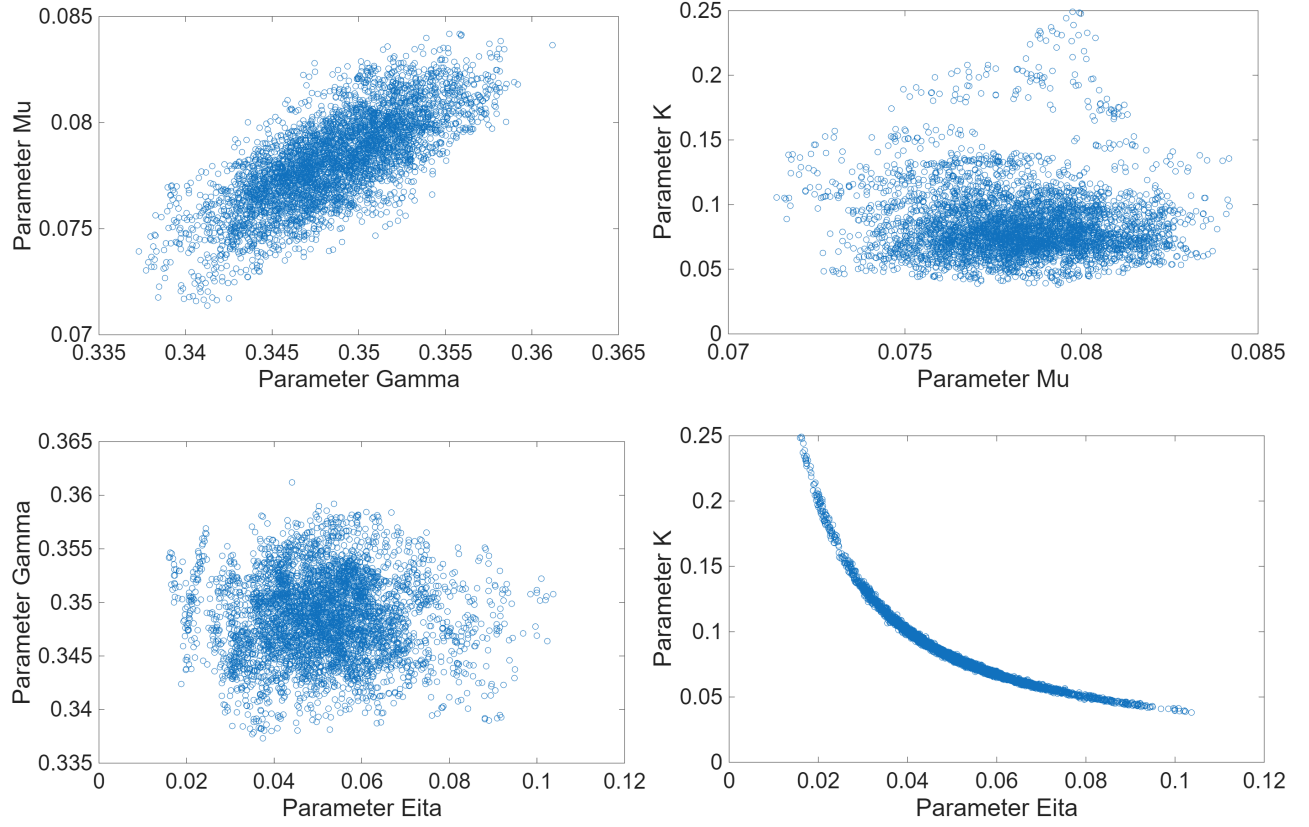


Figure 15 compares the marginal densities of DRAM and MCMC for each parameter. We note that they are similar, with the MCMC plots being slightly more closely distributed, which follows from the smaller estimated variance.

Figure 16 shows the pairwise plots by DRAM and MCMC. We note the similarity. We see from the pairwise plots that the parameters are highly correlated, and so global sensitivity analysis techniques that rely on independence would not be applicable. Making that assumption in this situation would be inappropriate, and likely lead to unhelpful results.

Figure 11: 2c Chains under DRAM

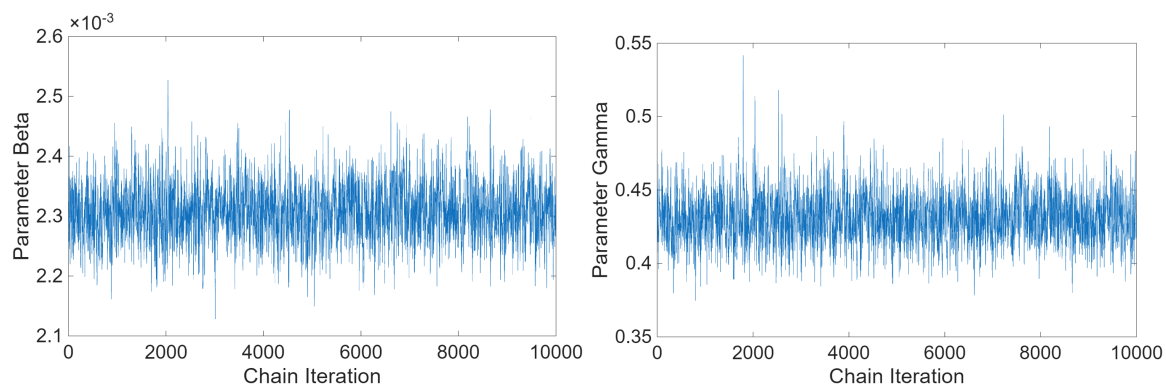


Figure 12: 2c Distributions under DRAM

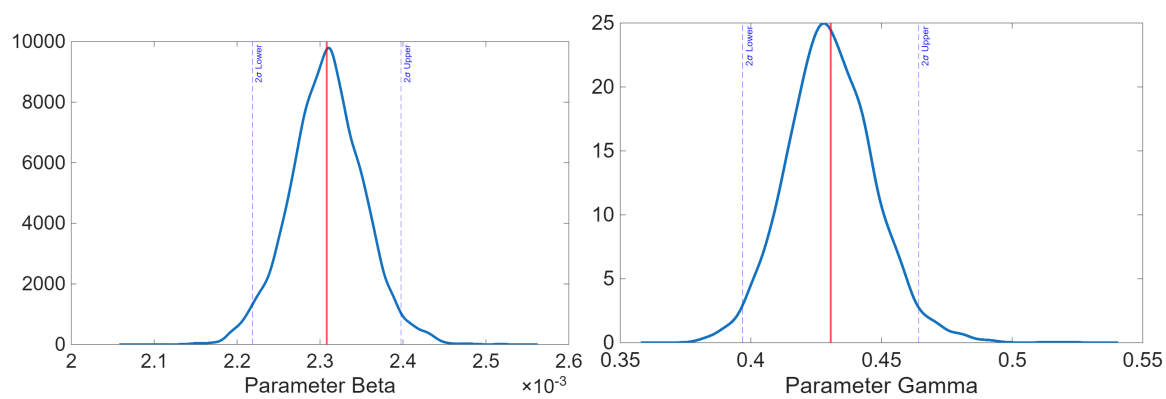


Figure 13: 2c Pairwise Plot under DRAM

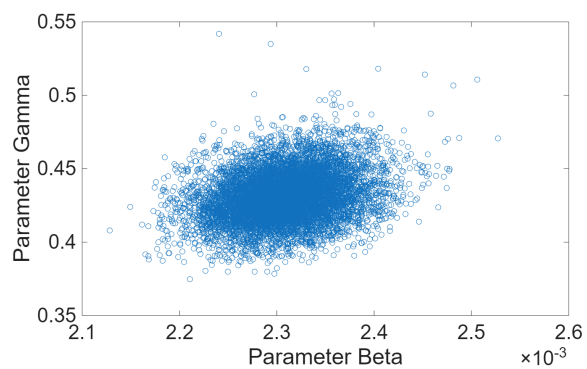


Figure 14: 3 Chains under DRAM (Left) and MCMC (Right)

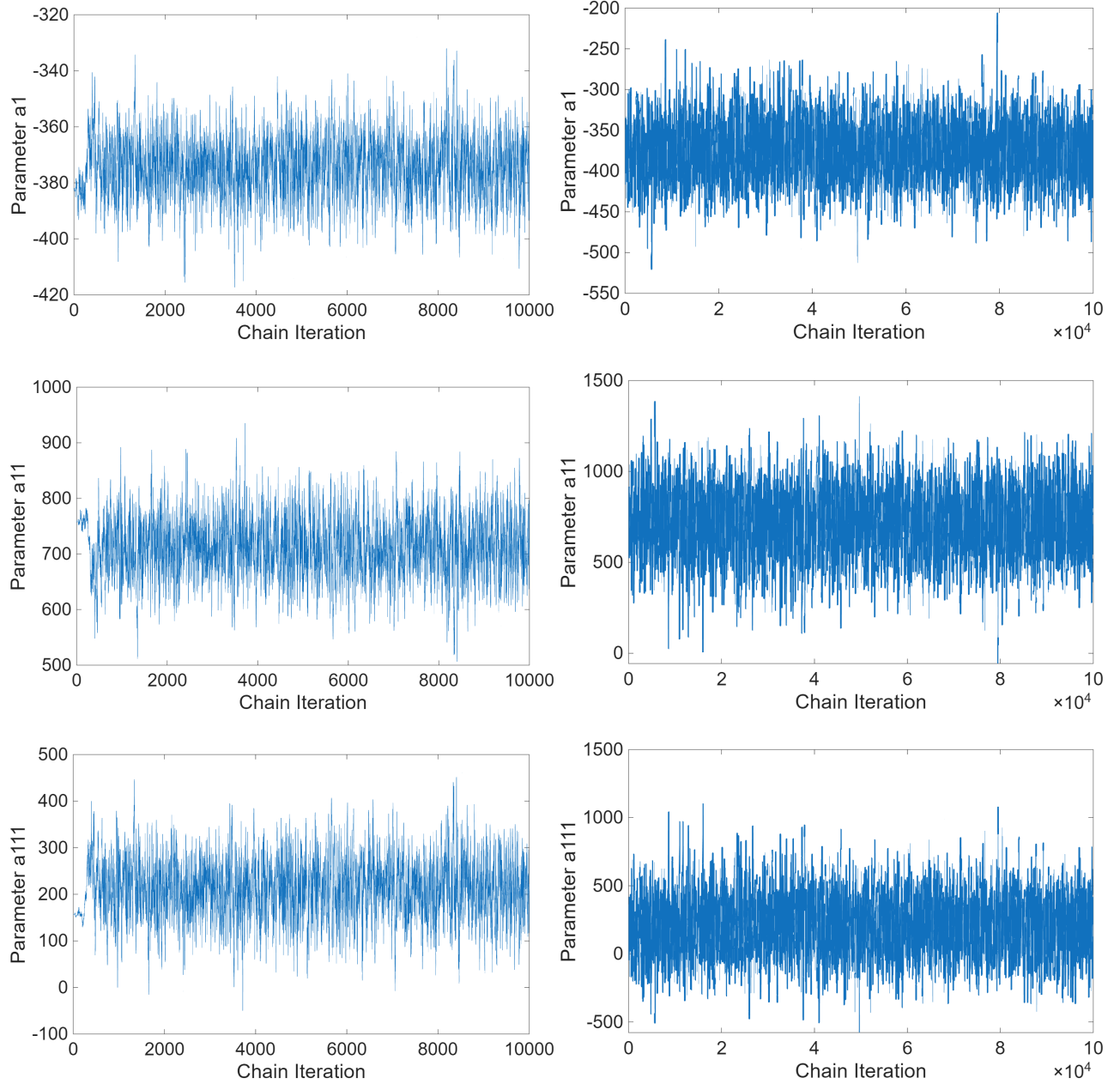


Figure 15: 3 Densities under DRAM (Left) and MCMC (Right)

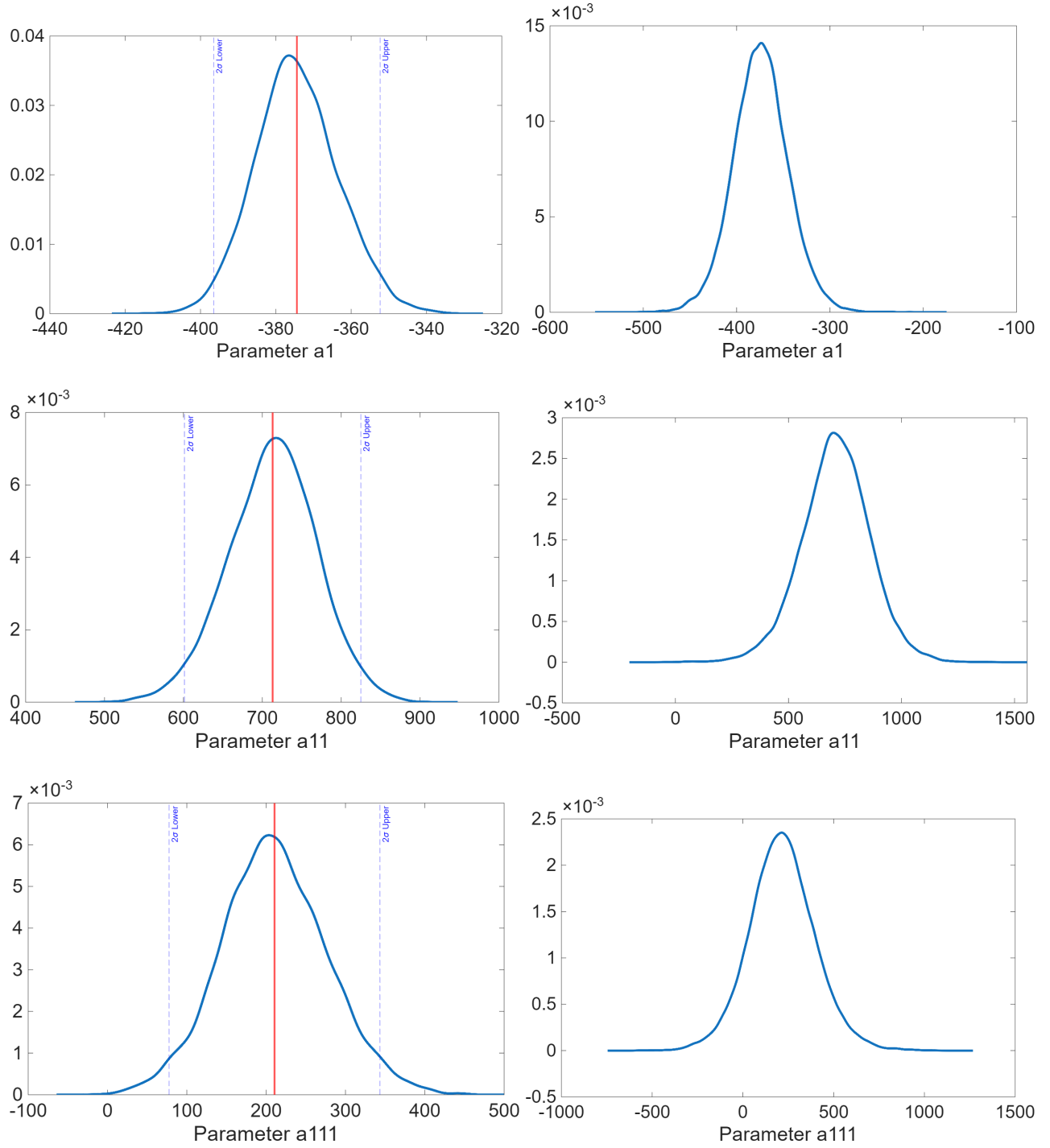


Figure 16: 3 Pairwise by DRAM (Left) and MCMC (Right)

